

Various Models of Thermal Imagers

Parameter	FLIR		Fluke		Optotherm		Avio (NEC)		Testo	
	FLIR A35 sc / A15 sc / A5 sc	FLIR SC5000	Fluke Ti50FT	Fluke Ti55FT	Opto-Therm EL	Opto-Therm Micro	Avio (NEC) InfReC R300 SR	Avio (NEC) InfReC H2640	testo 890	testo 885
Field of view	48° (H) x 39° (V) with 9mm lens 44° (H) x 36° (V) with 5mm lens	42° (H) x 34° (V) to 10° (H) x 8° (V)	23° (H) x 17° (V)	23° (H) x 17° (V)	50° (H) x 37.5° (V)	50° (H) x 37.5° (V)	22° (H) x 17° (V)	22° (H) x 17° (V)	42° (H) x 32° (V)	30° (H) x 23° (V)
Standard temperature range	-40°C to +160°C	5°C - 300°C	-20°C to 350°C	-20°C to 600°C	0-300°C	0-300°C	-40°C to 500°C	-40°C to 500°C	-20°C to 100°C / 0 - 350 °C (switchable)	-20°C to 100°C / 0 - 350 °C (switchable)
Resolution	320 x 256 or 160 x 128 or 80 x 64	640x512 or 320x256	320 x 240	320 x 240	320 x 240	320 x 240	640 x 480	640 x 480	640 x 480	320 x 240
Sensitivity (NETD)	< 50 mK	< 20 mK	< 70 mK	< 50 mK	< 50 mK	< 200 mK (microscopic lens) < 50 mK (wide angle lens)	< 25 mK	< 60mK (25mK S/N improvement)	< 40 mK	< 30 mK
Accuracy	±5°C or ±5%	±1°C or ±1%	±2°C or ±2%	±2°C or ±2%	±2°C or ±2%	±2 °C or ±2%	±1°C or ±1%	±2 °C or ±2%	±2 °C or ±2%	±2 °C or ±2%
Spectral response	7.5-13µm	2.5-5.1µm	8 - 14µm	8 - 14µm	7-14µm	7-14µm	8 - 14µm	8 - 13µm	8 - 14µm	8 - 14µm
Power supply	12/24V DC	12V DC	Two rechargeable Li-Ion smart batteries (operate 2-3 hours)	Two rechargeable Li-Ion smart batteries (operate 2-3 hours) and 110/220V AC, 50/60Hz AC adaptor	100-240 V AC, 50/60Hz, 4A	12V DC	Li-Ion battery (operates 2-3 hours) and 100-220V AC, 50/60Hz	Li-Ion battery (operates 2-3 hours) and 100-220V AC, 50/60Hz	Two rechargeable Li-Ion batteries (operate 4.5 hours)	Two rechargeable Li-Ion batteries (operate 4.5 hours)
Mount	Goose-neck mount	Fixed	Hand-held or tripod mount	Hand-held or tripod mount	Fixed	Fixed	Hand-held	Hand-held	Hand-held or tripod mount	Hand-held or tripod mount
Size (L x W x H)	106x40x43mm	320x141x159mm	162 x 262 x 101mm	162 x 262 x 101mm	690 x 690 x 1140mm	89 x 76 x 200mm	121 x 105 x 193mm	110 x 110 x 210mm	253 x 132 x 111mm	253 x 132 x 111mm
Weight	0.2kg	3.8kg	1.95kg	1.95kg	73kg	12kg	1.3kg	1.7kg	1.63gm	1.57gm

imager, and then we overlayed and blended the thermal image with the visual image. This enabled the user to exactly identify the correct location of the issue identified by the thermal imaging camera," says Thakkar.

For instance, some of the new uncooled systems that are available over the past year have seen some interesting functionality added to them.

"Wi-Fi connectivity for data transfer through iPhone/iPad or Android devices, as well as Bluetooth connectivity for voice annotations and current readings. Touch-screen displays, intuitive user interfaces (UIs) and excellent ergonomics along with tiltable IR units are also seen," adds Singh.

Thermal imagers for electrical applications are used in harsh envi-

ronments so they need to be rugged enough to withstand the environment and serve for a long period of time.

Cooled or uncooled detectors

Different types of detectors have different thermal and spectral sensitivities. In addition, they have different cost structures due to various degrees of